A New Solution for Fault Localization of Reversible Circuits

Ming-Cui Li, Ri-Gui Zhou

College of information school, East China Jiaotong University, Nanchang 330013, China
cacaony@gmail.com

***Abstract*—Fault detection and location are important steps in the design and production of reversible circuits. However, there is little research work has been done on fault location for reversible circuits. In this paper, two algorithms labeled LRCMD and MSDA are proposed to locate the single missing-gate fault into a small module. LRCMD divides the circuit into several large models from the lateral based on the dependency relationships of the gates in reversible circuits. MSDA conducts secondary division and adjustment for the result of LRCMD from the longitudinal according to the dominated relationship between gates. These two algorithms provide a new solution for single missing-gate fault location which has never been studied before. Simulation results show that the algorithms yield very good performance.**

Keywords-Reversible Circuit; Fault Detection; Fault Location; Quantum

I. INTRODUCTION

Reversible circuit proposed by Landauer [1] and Bennett [2] is becoming a promising technology in quantum computing, nanotechnology and low power CMOS. A lot of research work about reversible circuits has been done by scholars[3-5]. To guarantee the correctness and durability of reversible circuits, fault detection and analysis are very important in the phase of design and production as well as after that. Fault models proposed by scholars are mainly divided into two categories which are stuck at fault model and gate fault model. Stuck at fault model refers to the model that one or more points of the wires in the circuit are stuck with some fixed logic value, 0 (s-a-0) or 1 (s-a-1)[6]. Since the stuck at fault model does not take into account whether the reversible gates work correctly, it is a kind of passive fault detection method. Moreover because of the large number of possible fault points, algorithm of generating test vectors for this kind of model is rather complex. Gate fault model refers to some gates do not work correctly, it is a kind of active fault detection model, and can better reflect whether the function of the circuits has implemented correctly.

Several gate fault models have been proposed including: single missing-gate fault (SMGF), repeated-gate fault (RGF), multiple missing-gate fault (MMGF) [7] and partial missing-gate fault (PMGF)[8]. John P.Hayes [9] put forward DFT method to detect SMGF fault. An extra wire and a number of controlled-NOT gates are added to the original circuit in DFT. The advantage of DFT is that it can detect any single missing-gate fault in the circuit by just using one test vector. But the disadvantage of DFT is also obvious: the redundant controlled-NOT gates added to assist the fault detection themselves may also generate new faults, and once new faults occur, the advantages of DFT method will not exist. Furthermore DFT method cannot locate the fault. Xiao Fangying etc. [10] proposed CTS algorithm to detect SMGF. CTS also can not locate the fault and only applies to small reversible circuits. There is no gate fault localization method has been put forward yet. Especially, for the large complex irregular reversible circuits, no feasible gate fault detection and localization method has been found yet. This paper will put forward two modular division methods LRCMD and MSDA to detect as well as locate SMGF fault for complex irregular reversible circuits .

II. GATE MISSING FAULT DETECTION

A reversible circuit is composed of reversible gates. Among the reversible gate libraries, NCT library and GT library are the most commonly used and studied. NCT library consists of 3 basic gates: NOT gate, controlled not gate (FG), Toffoli gate (TG). These gates are shown in figure 1, where $\bullet$ represents the control bit, and $\oplus$ denotes the target bit. NOT Gate is a 1×1 quantum gate, whose function is to invert the input. The controlled not gate implements the logical function: $|A,B\rangle \rightarrow |A, A\oplus B\rangle$. When the control bit A is set to $|1\rangle$, the target bit B will be reversed, otherwise B will remain unchanged. The function of Toffoli gate is $|A,B,C\rangle \rightarrow |A,B,AB\oplus C\rangle$. When the two control bits A and B are both $|1\rangle$, the target bit C will be flipped. GT library includes the gates whose structures are the extensions of Toffoli gate. The extensions of Toffoli gate are n-bit Toffoli gates with the function of $|X_1,X_2,...,X_n\rangle \rightarrow |X_1,X_2,...,X_{n-1},(X_1X_2...X_{n-1})\oplus X_n\rangle$.

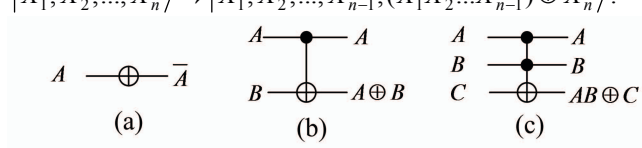

Figure 1. (a) NOT gate; (b) CNOT gate; (c) Toffoli gate

SMGF base on NCT and GT library is defined as a complete disappearance of one gate from the circuit, as shown in figure 2. In a NCT-based or GT-based reversible circuits, all single missing-gate fault are detectable and localizable and the number of the test vectors is not more than the number of the gates. Because the one-to-one correspondence between the input and the output of reversible circuits, each gate can be detected by the vector

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$|111\ldots111x\rangle$ at the stage of the gate, where $|1\rangle$ acts on the control bit, and $|x\rangle$ on the target bit, $|x\rangle$ may be $|0\rangle$ or $|1\rangle$. The original input test vector of this gate can be reverse calculated from $|111\ldots111x\rangle$. If there is one SMGF fault on gate a, due to the lack of transformation of the target bit, the correct output and a failure output will not be the same. So each gate in the circuit can be detected with an corresponding input vector, the complete test set size for SMGF is not more than the number of gates.

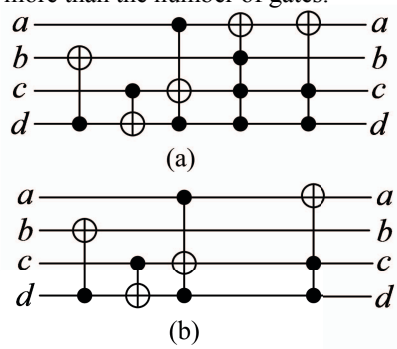

Figure 2. Example of single gate missing fault model. (a) is the original circuit, (b) is the circuit after missing the fourth gate of (a)

III. LRCMD(Large Reversible Circuit Module Division) Algorithm

The goal of algorithm LRCMD is to get a lateral disjoint module set $S=(M_1,M_2,\ldots,M_r)$ for a given reversible circuit. In LRCMD, each gate is characterized by a 2-tuple $G\{w,d\}$, w represents the wires occupied by the gate with the target bit placed on the last wire, d is the depth of the gate. We use $w(g_i)$ to represent the wires occupied by gate g_i, as shown in figure 3, $w(g_9)=\{b_0,a_0,a_1\}$.

Gate dependency refers to the inputs of the control points of a gate are the output of some gates. We use $g_p \mapsto g_q$ to represent gate g_p depends directly on the reversible gate g_q. Gate g_{10} dependent directly on g_1 and g_9 in figure 3.

The logical depth of a reversible gate refers to the column number the gate located in the circuit according to the execution order from left to right. Expression $d(g_i)$ denotes the depth of gate g_i. For instance, the value of $d(g_9)$ is 6 as shown in figure 3.

Algorithm LRCMD(large reversible circuit module division):

1) Initializing one set V_s which contains all the gates of the reversible circuit, and are sorted according to the depth of each gate.
2) Remove the first gate g_k from V_s, start building module M_i from gate g_k.
3) Search for the next gate g_n by the least-depth-first rule. If $d(g_n)\geq d(g_r)$ and $w(g_n)\cap w(M_i)\neq\varnothing$, wherein $d(g_r)$ is the depth of the last gate added in the current module M_i, start the reverse search according to the dependence of the control points of the gate. If the dependencies of g_n ($g_n \mapsto g_p \mapsto \ldots \mapsto g_k$) satisfies at least one of the following conditions, add g_n into module M_i, and remove it from V_s.
   a) All the control points of g_n reverse depends on the gates in module M_i.
   b) Some of the control points of g_n reverse depend on the gates in module M_i and others directly reverse depend on the input.
4) The module division completes if one of the following conditions is met:
   - All direct subsequent gates of the gates in module M_i are searched.
   - The subsequent gates have joined a module.
   - All gates in V_s are retrieved.
5) Take the first gate g_i from the remaining gates in V_s to start the next module M_j, neglecting the gates which have added into a module when searching gates for the next module M_j. Constructing M_j according to step 3, 4, 5.
6) Repeat step 5, until V_s is empty.

LRCMD divides the relevant gates as a group from the input side, which is a horizontal division according to the functional dependence of each gates. We detect the modules in the sequence of division $M_1 \to M_2 \to \cdots M_i \to \cdots$, and stop detecting till the fault is found or all modules have been detected. When module M_i is being detecting, the directly input of all the modules after M_i, namely, $M_{i+1},M_{i+2},\cdots$, are set to $|0\rangle$, then all the controlled gates after M_i will not work, so the test vector can not detect any controlled gates after module M_i. Because there is no fault has been detected in modules $M_1 \to M_1 \to \cdots \to M_{i-1}$, if the corresponding output of the test vector for M_i is wrong, the fault must exist in module M_i.

According to LRCMD, the 5 bit adder is divided into four modules $\{M_1,M_2,M_3,M_4\}$ as shown in figure 3. Deferent modules are separated by the dotted lines. The circuit is detected by the order $M_1 \to M_2 \to M_3 \to M_4$. When detecting module M_1, we can use the test vector $|11010000000\rangle$, which can make the controlled gates in M_1 work only. Because all the control bit of gates in module M_2,M_3,M_4 are set to $|0\rangle$, these gates will not work. If there is SMGF fault detected by vector $|11010000000\rangle$, the fault is certainly in M_1. Similarly, test

vector $|00000100000\rangle$ affects the gates in M_1 or M_2 only, without affecting the gates in M_3 or M_4.

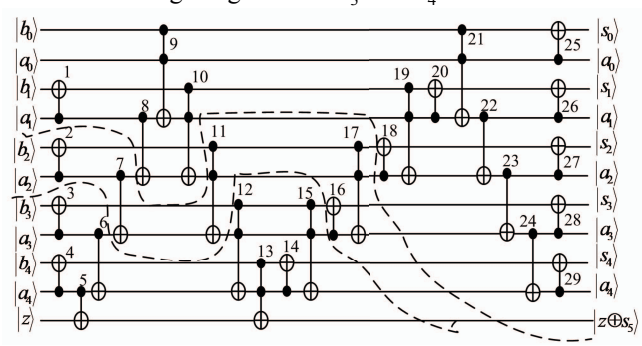

Figure 3. large module division of 5 bit adder

IV. MODULE SECONDARY DIVISION AND ADJUSTMENT

Not all the modules in LRCMD can be detected by only one vector. For instance, test vector $|11010000000\rangle$ can not make g_{10} work in figure 3. So it is essential to secondary division and adjustment for modules achieved in section Ⅲ. According to whether the gates can be detected by the same test vector, the relationship between two gates in the circuit can be divided into two categories, independence and domination. Independent relationship refers to that we can not use the same test vector to detect these two gates, and domination means we can use the same test vector to detect them. One large module can be divided into some smaller simple modules according to domination relationship, and each smaller module can be detected only use one test vector. Secondary module division is a further partition algorithm on the result of LRCMD, and will divide the large modules from the longitudinal.

Algorithm MSDA (module secondary division and adjustment):

1) Calculate the direct input wires of the current module $wp(M_i)$ and that of the modules before the current module $wp(M_i, M_{i-1}, \cdots, M_1)$. For instance, the direct input wires of the second large module in 5bitadder shown in figure 3 can be represented as $wp(M_2) = \{b_2, a_2\}$, while that of the modules before module 2 (including module 2) is $wp(M_2, M_1) = \{b_0, a_0, b_1, a_1, b_2, a_2\}$.
2) Select the first gate g_k ($w(g_k) \subseteq wp(M_i, M_{i-1}, \cdots, M_1)$) which has not joined any small module from M_i as the first gate of the new small module M_{ij}, i is the number of large module, j indicates the jth small module in the ith large module.
3) Search for the next gate g_h. We use c_{hl} to denote the lth control point of gate g_h, and $w(c_{hl})$ denotes the wire occupied by the lth control point of gate g_h. If the number of target points between c_{hl} and the one before c_{hl} in the same line is even or there is not any control point before c_{hl} yet, and gate g_h does not meet any of the following cases, g_h will be added to module M_{ij}.
   a) $w(c_{hl}) \notin wp(M_i, M_{i-1}, \cdots, M_1)$, the number of control points and target points belong to module M_{ij} before c_{hl} are zero and even (greater than 0) respectively.
   b) c_{hl} Meets the condition: $w(c_{hl}) \notin wp(M_i, M_{i-1}, \cdots, M_1)$, and $w(c_{hl}) > max(w(M_{ij}))$.
4) Expand the original wires of each small module to that of the large module it belongs to, namely, $wp(M_{ij}) = wp(M_i)$. Repeat step 2, 3, until all the gates meet the condition $w(g_k) \subseteq w(M_i, M_{i-1}, \cdots, M_1)$ have joined a small module.
5) Move the rest gates not belong to any small module from module M_i to M_{i+1}.

In step 2, as the beginning gate of a new small module, the wires occupied by the gate should be all included in the original input wires of the current and the previous large modules, for the sake of ensuring the test vector for this gate can not affect the gates in the later large modules. In step 3, on the basis of satisfying the domination relationship, two kind of gates which can not be detected are excluded. The first one is that some control point of a gate is the first control point on that wire of the current small module, but there are even (greater than 0) target points in front of it. According to our test vector generating rule, this control point will never be set to $|1\rangle$ in the current small module. Therefore, this gate can not be detected. The second is that the wire occupied by some control point of a gate is not included in that of the current small module and is not in the wires of the direct input wires of the current and the previous large modules, it is more suitable to move this gate to the next large module to detect. Because the test vector for this gate may probably make some gates on its left side belong to later large modules work.

In the 5 bit adder, gate g_1 is the start gate for small module M_{11}. The first small module of M_1 generated by MSDA is: $M_{11} = \{g_1, g_8, g_9, g_{18}, g_{21}, g_{22}, g_{25}, g_{26}\}$. Starting construction M_{12} from the first gate not added into M_{11} in the rest gates of M_1, we will get $M_{12} = \{g_{10}, g_{19}, g_{20}\}$. None of the rest gates not join M_{11} or M_{12} meets the condition of step 2 in MSDA, so no more small module of M_1 can be constructed. The gates not added into any small module in M_1 will be moved into the next large module M_2. The final large modules of 5bitadder are adjusted as shown in figure 4. The result of small module division is that

M_1 is divided into two small modules: M_{11}, M_{12}. And M_2, M_3, M_4 has only one small module respectively.

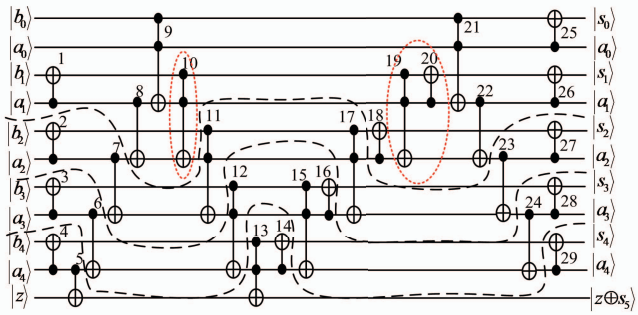

Figure 4. Secondary model division and adjustment for 5 bit adder

Each small module got by MSDA will be tested by the same test vector. We generate the test vector in this way: firstly, at the entrance of the small module, set each line to $|1\rangle$ if the first one on it is a control point, all other lines belonged to this or other large modules are set to $|0\rangle$; secondly, compute the corresponding test vector in the input side. For instance, the test vectors for M_{11}, M_{12} are $|11010000000\rangle$, $|00010000000\rangle$ respectively.

TABLE I EXPERIMENTS OF MODULE DIVISION USING LRCMD AND MSD

circuits	gates	large modules (result of LRCMD)	small modules (result of MSDA)
4b15g_1	15	1	4
5bitadder	29	4	{2,1,1,1}
8bitadder	122	4	{2,3,4,3}
Cycle17_3	48	1	3
Ham15	132	1	10
Hwb4	17	1	2
Hamb6	126	1	14
4bitcomparator	35	4	{3,2,1,1}
Ramdom1	86	2	{8,7}
Ramdom2	128	3	{7,8,7}
4_49tc1	16	1	3

TABLE II COMPARISON OF DFT,CTS AND MSDA

circuit	gatecount	wirecount	DFT	CTS	MSDA
4b15g_1	15	4	5	4	4
5bitadder	29	11	2	3	5\2
8bitadder	122	24	13	6	12\5
Cycle17_3	48	20	29	3	3\3
Ham15	132	15	47	8	10\10
Hamb4tc	17	4	7	2	2\2
Hamb6tc	126	6	83	19	14\14
4bitcomparator	35	16	4	2	7\2
Ramdom1	86	16	53	8	15\8
Ramdom2	128	24	74	9	22\9
4_49tc1	16	4	7	4	3\3

We conducted a set of experiments to examine the algorithms. All the reversible circuits tested are from the web site of Reversible Logic Synthesis Benchmarks Page (http://webhome.cs.uvic.ca/~dmaslov/). The experimental results are shown in Table I and table II. In table II, the left result of MSDA is gotten by first divided by LRCMD, and then adjusted by MSDA. The right one is the result of using MSDA only. Comparing to other two algorithms DFT and CTS, our algorithms not only can detect SMGF fault, but also can locate the fault into a small module. Furthermore, the test set for LRCMD and MSDA is much more smaller than that of DFT, and is approximately equals to that of CTS as shown in table II. All the ultimate purpose of detecting faults is to correct faults, and fault location is an essential step. LRCMD and MSDA is one step closer to our final goal, and they yield very good performance for the complex reversible circuit .

V. CONCLUSIONS

In this paper we investigated the properties of NCT-based and GT-based reversible circuits under the single missing-gate fault model (SMGF). Based on the properties we derived, two efficient module partition algorithms LRCMD and MSDA are proposed to detect and locate SMGF fault. LRCMD is put forward for large module division from horizontal according to the gate dependency relationship in ascending depth. MSDA is a vertical subdivision and horizontal adjustment algorithm on the results of LRCMD. One large module can be divided into several smaller modules by MSDA, and each small module can be detected by only one test vector. We detect SMGF fault in the order of module generated. Each test vector we generate only influence the gates of the current module and before, and will never make any gate work in the modules after the current module, so the algorithms we put forward can also locate SMGF fault into a small module which has not been studied before.

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